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Prefabricated biogenic construction and their fire-safety properties – a literary review

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ABSTRACT: This paper presents initial findings from the multi-disciplinary research project: Prefabricated Biogenic Building – construction, fire and architecture (2024-2026) that includes fire-safety technicians, contractors, thatchers, and architectural researchers. The project builds upon the former project: Thatched Facades for the Green Transition (2021-2023) that examined fire-retardant properties of clay built into thatched constructions. The current project investigates the challenges linked to developing biogenic prefabricated constructions for multi-storey buildings, focusing on buildability, fire-safety and classification of specific materials/components, and architectural quality. The project's initial phase examines state-of-the-art. By using Structured Literature Review (SLR) biogenic construction materials and components are investigated from two angles: 1) existing knowledge regarding prefabricated constructions and building techniques 2) fire-safety. Unfortunately, it shows that very little R&D is made despite the great need for knowledge about renewable constructions in architecture. Concludingly, three façade design proposals are presented responding to findings made in the literature review.

1 INTRODUCTION

Several political initiatives in Denmark based on international analyses of the climate crisis and environmental crisis is presently forming a new set of rules and guiding principles aiming at the Danish building sector. In 2020 the Danish Government introduced the Climate Act, which is the political ambition that Denmark must reduce the greenhouse gas emissions by 70 percent in 2030 compared to the level in 1990. This on order to reach climate-neutrality of the Danish society by 2050 (Retsinformation, 2020). The same year the 'Voluntary Sustainability Class' [FBK] was launched as an addition to the Danish Building Regulations which is targeting 8 kg CO_2/m^2/per year. This goal is deemed to be sustainable (BPST, 2021). Concurrently, the Danish Non-profit Association Reduction Roadmap has introduced the 'Reduction Roadmap 2.0' suggesting a maximum of 5.8 kg CO_2/m^2/per year (Reduction Roadmap, 2023). 'The Reduction Roadmap 2.0 is an Earth-shot project which works to translate the Paris Agreement and the Plan-etary Boundaries for Climate Change into industry-specific CO_2 reduction targets for over 90 percent of Danish new build projects. Finally, according to the scope of the Planetary Boundaries developed by the Stockholm Resilience Centre the reduction numbers needed are closer to 0,4 kg CO_2/m2/per year (Rockströmn et al. 2009).

At present the Danish construction industry is accountable for 30 percent of the total CO_2 emissions at national level – 20 percent comes from buildings (heating/electricity) and 10 percent from materials and construction. Pioneering Danish research into applied building technology suggests that significant reductions can be found in an increased use of biogenic building materials across the construction industry and at all levels (Rasmussen, 2022. Beim et al.,

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2023. Hau et al, 2024). With reference to the critical aspects listed above linked to climate crises, the Danish Building sectors CO_2 emission problem and the Danish government's political ambitions dealing with mitigating the problem - the promising reductions that can be reached by changing the building industry, leads to questions as: How can we speed up the processes without losing the local building culture? How do we spark (and keep) the engagement of the building industry? How can we bring different sorts of technical knowledge, skills and interest together? How can biogenic prefabricated construction solutions, for single housing as well as multi-storey buildings be implemented – to reduce the negative impact of construction on the climate and the environmental conditions?

This paper presents the initial findings of the multi-disciplinary research project. The Prefabri-cated Biogenic Building – Construction, Fire and Architecture (2024-2026). The disciplines involved in the project are fire engineering from DBI (The Danish Institute of Fire & Security Technology), contractors with competencies in prefabricated building from Enemærke & Petersen, and EcoCocon/Denmark, specialized craftmanship of thatching from Tækkefirmaet Horneby and Hemmed Tækkefirma, and architectural research represented by CINARK, Institute of Architecture and Technology, The Royal Danish Academy. The project is partly funded by The Environmental Technology, Development & Demonstration Program (MUDP), also called the Danish Eco-Innovation Programme. The MUDP programme supports projects that; 'deliver real environmental improvements, while at the same time strengthening Danish enterprises' and 'green competitive capabilities.'

Through literature reviews the state-of-the-art in Europe has been mapped. Here existing knowledge concerning prefabricated biogenic constructions, relevant building techniques, and biogenic materials with a focus on their fire-safety properties have been examined. The paper discusses the findings, the challenges, and possible solutions related to prefabricated biogenic facade constructions for multi-storey buildings based on state-of-the-art and three designs proposals.

Why dealing with thatch? Fire-safety is key when aiming at developing sustainable architectural solutions with biogenic building materials. Thatch is part of the Danish building traditions and today the Danish Thatches Guild amongst other parties in the building industry want to expand the use of thatch not only for roofing but also for facades to improve the negative CO_2 footprint from construction. To make this happen research must be made.

2 METHODOLOGY

2.1 *Defining search methods for the literature reviews*

When setting out to conduct literature reviews focusing on topics as; the use of biogenic materials in façade construction, fire safety techniques found in the craft of thatching, and new fire-testing regimes for biogenic materials and constructions, qualified speculations founded in existing research and built demonstrators related to the field, were to be found in the greater pool of knowledge around the world. It was expected to find various sorts of literature, that presented experiments and research into built projects using biogenic materials as reed or straw in new ways. Also, projects applying locally produced biogenic materials through sustainable practices, that are inspired by vernacular traditions or demonstrating simple construction methods developed by craftsmen or local entrepreneurs for the sake of sustainable business development. However, the findings turned out quite differently.

To specify and overcome the task, the literature review was decided to be narrowed down - based on the method: Structured Literature Review (SLR), which is designed to secure a transparent and reproducible process, minimizing biases by providing an audited trail of reviewer's procedures, decisions and conclusions (Tranfield et al. 2003, Grant & Booth, 2009).

first part of the literature review concerning 'technical aspects of fire' focuses on selected biogenic materials, building components, and their 'fire technical performance' described within the context of Danish Building Regulations and EU standards (pre-accepted solutions) as well as emphasizing the practical applications of fire-safety engineering and testing methods (performance-based design). Furthermore, it describes how these properties have implications for the

design task of the architects, fire-safety consultants, and producers of biobased building products, when planning buildings up to three floors regulated by fire conditions. The second part looking into prefabricated construction, was also narrowed down to focusing on fast-growing, biogenic materials, such as hemp, straw, eelgrass, and reed. These construction materials were considered in combination with load-bearing structures of other biogenic materials as timber, which is used, developed, and tested in relation to 'fire for assembly' construction in Europe. Furthermore, three Danish experimental studies: a) Thatched facades for the green transition (2021-2023), b.) Application of Biogenic Materials in Construction (2021-2022), and c.) 'Dwellings 4to1 initiative' (2022-2023) were analyzed separately, as the latter two projects happen to build upon findings and results from the first project, which also forms the groundwork project for the present research project. (Beim et al. 2023, Rasmussen et al. 2022, Realdania 2023).

Table 1. Diagram of the protocol for SLR.

Database	Search Query	Tweeks	Hits	Relevant hits
Proquest: https://www.proquest.com/	A and B*	None	46	0
ScienceDirect: https://www.sciencedirect.com/	A and B*	Environmental Science; Engineering; Energy; Material Science; Social Sciences	4.446	48
Architecture, Design and Conservation: https://adk.elsevierpure.com/	A and B*	None	101	4
DTU: https://orbit.dtu.dk/en/	A and B*	None	36	0
Aarhus Universitet: pure.au.dk	A and B*	None	24	0
SDU: https://www.sdu.dk/en/search	A and B*	None	8	0
KU: https://forskning.ku.dk/soeg/	Fail – database error	None		
Aalborg: https://vbn.aau.dk/	A and B*	None	654	6
SDU, Aarhus University and Aalborg	Eelgrass	None	11	3

* A-search query: (prefabricate OR modular OR assembly) AND (straw OR reed OR thatch OR grass OR hemp) AND fire. B-search query: (prefabricate OR modular OR assembly) AND (eelgrass) AND fire

Several knowledge gaps on various technical aspects were found e.g. the durability of the thatch in relation to the gradient of a facade/roof and the buildability of modular thatched elements. To fill these gaps organizations of thatchers across European countries, as well as several patents on thatched elements were reviewed.

Relevant databases have been specified according to both the architectural, constructional, and fire-technical context (see Table 1). The scope of the inquiry has been defined as biogenic building materials for 'montage construction' [assembly construction] in Europe, which is developed, in use and/or have been tested in regard of fire-safety. The term 'Montage' was chosen to be applied since it is a European widely used term or concept referring to industrialized, prefabricated building components or modules designed to be assembled on site (Gravesen, 2011).

The SLR has focused on fast grown biogenic plant materials such as: hemp, reed, straw, grass, eelgrass possibly in combination with a framework structure made of slow grown plant materials as wood. Exotic species as bamboo have not been included, although these sorts of plants could be relevant in future scenarios. Yet, these species can have a negative impact on endemic flora and fauna and therefore must be grown with care. Fungi is also not included, as these species hold a different biological classification than plants. While cork was not initially

included in the study, more articles on the subject occurred during the database search. Cork was therefore included in the study, as it is an endemic species in Europe as well as e.g. Portugal has a well-established production of cork materials and products.

The SLR was carried out in the open-sourced reference system *Zotero* used for protocol and it was assessed in a double-blind process by two researchers. Furthermore, the 'Snowball' approach was used to search for relevant secondary literature. Snowball is a technique for localizing additional relevant literature through the primary literature found in the SLR (Wohlin, 2014). This concept was expanded to include studies, papers, and other sorts of literature provided by the consortium as well as anecdotal documentations from relevant craftsmen.

3 FINDINGS

3.1 *Fire properties of biogenic materials*

When designing buildings most countries and Denmark included use prescriptive 'Pre-Accepted Solutions' (PAS), which are a faster way of defining the construction design according to the fire-regulations, but often it is very rigid determining the final design (Bjelland & Njå 2022). The alternative 'Performance Based Design approach' (PBD) allows for a more innovative and project-specific fire-safety strategy compared to the prescriptive approach. PBD often requires a detailed understanding of fire dynamics and advanced testing/simulation methods (e.g. non-standard fire testing and evacuation modelling) making it a more complex approach, but it can also create a better potential for ensuring fire-safety in new building practices, such as building designs with biogenic materials.

Understanding fire-testing methods is crucial for evaluating the fire-safety of biogenic materials. Two general key types of tests have been highlighted during the review: reaction to fire tests and fire resistance tests. Reaction to fire tests evaluate the material/component contribution to the fire growth (in terms of heat release when burning) and smoke production. The European reaction-to-fire classification system for construction products except floors ranks materials and products in the Euro fire-classes A1 to F (best to worst performing). Most of the tested untreated biogenic building materials are ranked as fire-class E. For comparison, solid non-fire treated timber products are often ranked as class D.

Fire resistance tests (REI) evaluate the construction's ability to withstand fully developed fire conditions (represented by the so-called standard fire curve) without losing its structural integrity, load bearing capacity and construction's ability to restrict/insulate the heat flow through the construction. Due to their combustibility, biogenic materials represent unique challenges regarding fire-safety. The SLR found following consensus across the literature involved:

- Biogenic materials such as straw, reed, grass, hemp, and eelgrass generally exhibit poor reaction to fire properties when untreated. For example, untreated straw, reed, and grass typically achieve only Euro Class E [EM1], indicating a high susceptibility to ignition and fire spread. How-ever, with appropriate treatments, such as plastering (e.g. clay or lime) or chemical fire retardants, these materials can achieve higher classifications, such as Euro class B-s1d0, which significantly enhances their fire performance.
- In most studies the reported reaction-to-fire class for pure straw is E, see e.g. (FCBA 2009). If plastered with lime, earth, clay or covered with boards the reaction to fire class of the product (so-called "end-use") can reach to B -s1,d0 (Gaujard 2020, EcoCocon 2021). Straw insulation, if protected by clay, lime, reinforced cement/stucco, earth (26mm thickness) can achieve fire resistance classification even up to of REI-120. Straw insulation can also be protected with boards, e.g. gypsum plasterboard, OSB, (12.5 mm thickness) potentially reaching REI90 (The Last Straw 2024).
- Reed (thatching) is commonly used in construction for roofing and façade thatch. Reed (Phragmites australis) and (Miscanthus) exhibit critical heat fluxes for ignition between 10-15 kW/m². Untreated reed is rated class E. Fire retardants can improve this to class C-s2,d0 (Efectis 2023). Tests by the Royal Danish Academy and DBI show bio/earth-based treatments, such as clay coatings can improve reaction-to-fire performance in small scale

indicative testing (Mini SBI and SBI) though correlations with standard classification tests remain unverified (Beim et al. 2023).

- Grass is used in construction for thermal insulation, wall/ceiling panels as well as in traditional and modern green roofs.(Grass-based thermal insulation products are generally rated Euro Class E. In fire resistance tests, a construction with grass fiber insulation protected by chipboard and gypsum fiber board achieved an REI 60 rating, failing due to integrity at 64 minutes (DBI 2023).
- Industrial hemp, derived from (Cannabis Sativa), is used in construction for products like hempcrete, insulation, and fiberboards. Pure hemp insulation typically reaches Euro Class E for reaction to fire. Plastered hemp blocks and construction boards can achieve euro class B-s1,d0. Hemp blocks exhibit minimal flaming and heat of combustion. Hempcrete blocks composed of hemp and lime exhibited good fire resistance properties showing potential for passing 120-minute test without losing its structural integrity and insulation (Shewalul et al. 2023).
- Common eelgrass (Zostera Marina) traditionally used as a roofing material on Denmark's Læsø island and in China, is now being explored for façade and insulation applications. Flame retardant treated eelgrass insulation mats, with a density of 80-200 kg/m³ can achieve fire classification C-s1,d1. Despite the quick ignition, tests have shown lower heat release rates compared to wood-based mats (Kugo & Mai 2023).

3.2 *Prefabricated biogenic constructions in relation to fire*

The second part of the SLR showed that research on prefabricated biogenic construction (montage) that also have been subjected to fire-testing are sparse in the scientific databases. Literature on biogenic fast-growing materials primarily focuses on straw(bale) constructions, with several studies on moisture migration, load bearing and isolation capacity. Also, the study showed an overlap with growing research on textile production methods and fire-protection. The field of study was too broadly defined, which also became evident in the screening process. A large sum of the articles found was related to energy production, forest-, and nature restoration, agriculture and textile production. Nonetheless the study found that a lot of emerging research is oriented towards biogenic materials like jute, cotton, wool, flax, hemp, ramie, and sisals as a material component in composite materials/products together with inorganic materials.

The study did not localize specific products developed for biogenic assembly construction. This maybe can be explained by the circumstance that the product is not produced as an element for modular construction, but rather as a product without a specific application. Also, product development is often connected to a patents and research into these matters may not be available to the public. The SLR highlights following knowledge or gaps within prefabricated biogenic constructions:

- A study from Sun et al. (2023) points to a tendency towards prefabrication and industrialization of straw bale constructions. However, it is still in an early phase and a lack of quantitative knowledge on e.g. moisture migration under different manufacturing and context conditions is a barrier for upscaling. Furthermore, insufficient technical and practical knowledge is found in the design and construction phases relating to an industrial scale. The study points to EcoCocon as a leading manufacturer within the field, although they are still in the early stages of scaling (45.000 m2 of strawbale wall in 15 years (2008-2023)) (Sun et al. 2023).
- Positive results when applying plaster or other composite materials to the constructions to improve fire-resistance, however none with biogenic materials (Sun et al. 2023).
- Two trends can be highlighted: load-bearing construction with straw: a.) A platform-frame where each floor of a building works as a platform to support the next floor, and b.) 'dry-bale wall panels' - a load bearing frame in timber with compressed straw as insulation in the middle (Atelier Schmidt 2020; Beaudry et al., 2019; Cascone et al., 2018).
- Several examples of small-scale experimental buildings using biogenic materials have been found. One example is by FAZ architects, who built a two-story hangar in Meyrin, Switzerland. The construction of the building is in solid spruce with minimal use of nails or glue,

a wooden facade on the ground floor and a cork facade on the first floor (FAZ Architectes 2022). In addition, the study found few examples of multi-storey buildings in straw bale construction. One example is by Atelier Schmidt, which built a four-storey residential complex in Nänikon, Switzerland from prefabricated straw bale elements with a timber frame and lime plastered facades (Atelier Schmidt 2020).

- A patent on thatched roof-facade elements (easy-riet) showed earlier examples of prefabrication. The product consisted of a board with a layer of reed glued on top simulating a thatched roof. Furthermore, the patents on modular thatched facade elements showed an example of how to construct tight joints both vertically and horizontally (van 'tZand te Harderwijk 2014).
- 5 strategies for fire protection with biogenic construction were identified as having potential: fire impregnation with chemical treatment (MagmaFirestop 2017), lime plaster (Sun et al. 2023), or clay (Beim et al. 2023) etc., tightly thatched constructions that slows down the development of the fire due to lack of oxygen (Vakfederatie Rietdekkers 2024), sprinkler systems, and constructive protection like flame deflectors (Leimand & Schotanus 2024)
- The Netherlands have a well-established tradition for constructional standards, fire safety policies and fire testing regarding thatched facade. The constructional standards showed a positive correlation between the gradient of the roof and the longevity of the reeds, pointing to that a thatched roof with a 25-degree gradient lasted on average 5 years, 30 degrees 12 years, 40 degrees 28 years, and 50 degrees 40 years. Findings also showed multiple fire-safety assessments on thatched facades, where tests on densely thatched facades showed promising results in preventing fire from spreading uncontrollably (Vakfederatie Rietdekkers 2024).
- In Denmark few experiments have endeavored into industrializing biogenic constructions going from hand-held investigations assuming the scalability and artistic investigations into robotic fabrication (Buthke & Trempe 2016).

3.3 *Concluding remarks on the literature review*

The main findings in the SLR and snowball analyses on biogenic prefabricated constructions points to a primary focus on strawbale constructions. Innovation in the field seems to be driven by smaller companies, builders and enthusiasts more so than bigger actors in the building industry, which is a valuable and slow process. This also means that experimental biogenic construction is rarely built according to a modular system/assembly construction, but specifically for projects, often on a smaller scale. Biogenic materials generally exhibit poor reaction to fire properties, but with appropriate treatments, such as plastering (e.g. clay or lime) or chemical fire retardants, these materials can achieve higher classifications, such as Euro class B-s1d0. Furthermore, often used prescriptive fire testing approach is faster, but also more rigid which can be an obstacle for new materials and small innovation businesses. Introduction of biobased materials in construction may require employing performance-based design - a holistic fire safety engineering approach tailored to the objectives of specific projects. It may be required to mitigate the fire-hazards created by pure biogenic materials through informed architectural designs as well as passive and active fire protection systems [EM3]. Close collaboration between architects, fire-safety engineers, craftsmen with material experiences and material scientists is crucial for advancing safe and sustainable building practices.

4 PERSPECTIVES ON THE FOLLOWING RESEARCH PROCESS

Based on the findings from the literature reviews, three facade design proposals have been developed. They have been designed by use of two iteration loops including interviews with the project partners: DBI, craftsmen, and the manufactures. The interviews aimed to collect as many inputs as possible for the early design phase to secure an integrated design process, where knowledge regarding buildability and fire-technical properties are represented (Knudstrup 2004). This part of the project resulted in three initial design proposals:

I. A Dutch inspired construction method with a tightly thatched layers of reed. The intention is to secure a high-density and low levels of oxygen inside the construction, which

slows down the development of fire. It is presumed that the surface of the thatch will burn with small flames, and the fire does not enter deep into the structure.

II. Clay-impregnated thatch inspired by the project *Thatched facades for the green transition* (2023). Coarser reeds are used, and the thatch is mounted more loosely than in Test I.

III. Inspired by a traditional Germany technique, short stacks of reed are "glued" to a clay panel using clay adhesive. Each row of reed overlaps the next, protecting both the clay adhesive and the clay panel. The reed/clay panels are mounted directly onto an EcoCocon element. The thin layer of reed holds a limited amount of energy, presuming that fire development will be rapid but short-lived.

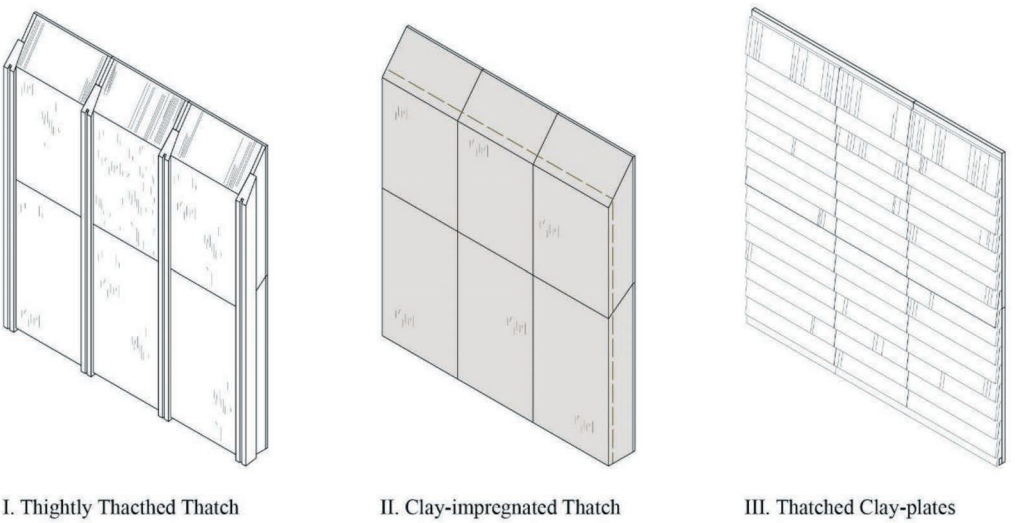


Figure 1. Sketches of the three façade designs.

In the next phases of the project; buildability, construction design, and fire-resistance performance of the three façade designs will be tested and the best performing façade design will be further developed to a full-scale multi storey facade construction for further fire-testing. At present it is the assumption that the fire-testing of the three façade designs will require flame deflectors to ensure so the fire cannot spread from floor to floor. The ambition is to develop biogenic architectural solutions that can meet the demands of future fire-safety regulations.

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